

Chap11-Exercises_LuisCorreia-745724_v1

November 23, 2025

1 MAP5935 - Statistical Learning (Chapter 11 - Survival Analysis)

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<https://www.statlearning.com/>

```
[1]: import numpy as np
import random
import pandas as pd
import matplotlib.pyplot as plt
from matplotlib.pyplot import subplots

from ISLP import load_data
from ISLP.models import ModelSpec as MS

from sklearn.linear_model import \
    (LinearRegression ,
     LogisticRegression ,
     Lasso)
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import KFold
from sklearn.pipeline import Pipeline

from sklearn.model_selection import \
    (train_test_split ,
     GridSearchCV)

from lifelines import KaplanMeierFitter, CoxPHFitter
```

1.1 *Conceptual Exercises*

- 1.1.1 (8) Recall that the survival function $S(t)$, the hazard function $h(t)$, and the density function $f(t)$ are defined in equations (1 – 3) below, respectively. Furthermore, define $F(t) = 1 - S(t)$.

$$f(t) = dF(t)/dt$$

$$S(t) = \exp \left(- \int_0^t h(u) du \right)$$

$$S(t) = \Pr(T > t) \tag{1}$$

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{\Pr(t < T \leq t + \Delta t | T > t)}{\Delta t} \tag{2}$$

$$f(t) = \lim_{\Delta t \rightarrow 0} \frac{\Pr(t < T \leq t + \Delta t)}{\Delta t} \tag{3}$$

Solution

(1) Showing $f(t) = \frac{dF(t)}{dt}$

For any $\Delta t > 0$,

$$F(t) = 1 - S(t) = 1 - \Pr(T > t) = \Pr(T \leq t)$$

Then we have

$$F(t + \Delta t) - F(t) = \Pr(T \leq t + \Delta t) - \Pr(T \leq t) = \Pr(t < T \leq t + \Delta t).$$

Therefore, using the definition of $f(t)$ in equation (3), we have:

$$f(t) = \lim_{\Delta t \rightarrow 0} \frac{\Pr(t < T \leq t + \Delta t)}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{F(t + \Delta t) - F(t)}{\Delta t} = \frac{dF(t)}{dt},$$

i.e. f is the derivative of the cdf F .

2. Expressing $S(t)$ in terms of the hazard $h(t)$

First note that, using the definition of conditional probability,

$$\Pr(t < T \leq t + \Delta t | T > t) = \frac{\Pr(t < T \leq t + \Delta t, T > t)}{\Pr(T > t)} = \frac{\Pr(t < T \leq t + \Delta t)}{S(t)}.$$

Plugging this into the definition of the hazard,

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{\Pr(t < T \leq t + \Delta t | T > t)}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\Pr(t < T \leq t + \Delta t)}{\Delta t S(t)} = \frac{f(t)}{S(t)}.$$

Thus

$$f(t) = h(t) S(t).$$

We also have $F(t) = 1 - S(t)$, so

$$\frac{dS(t)}{dt} = -\frac{dF(t)}{dt} = -f(t) = -h(t) S(t).$$

Hence S satisfies the differential equation

$$\frac{dS(t)}{dt} = -h(t) S(t), \quad S(0) = \Pr(T > 0) = 1.$$

Solving it by separation of variables:

$$\frac{dS(t)}{S(t)} = -h(t) dt \implies \int_{S(0)}^{S(t)} \frac{1}{s} ds = -\int_0^t h(u) du,$$

so

$$\log S(t) - \log S(0) = -\int_0^t h(u) du.$$

Since $S(0) = 1$, $\log S(0) = 0$, and we obtain

$$\log S(t) = -\int_0^t h(u) du \implies S(t) = \exp\left(-\int_0^t h(u) du\right).$$

1.2 Applied Exercises

1.2.1 (10) This exercise focuses on the brain tumor data, which is included in the ISLP library.

1.2.2 (a) Plot the Kaplan-Meier survival curve with ± 1 standard error bands, using the `KaplanMeierFitter()` estimator in the `lifelines` package.

```
[2]: # Load data
BrainCancer = load_data("BrainCancer")

durations = BrainCancer["time"].values
events = BrainCancer["status"].values    # 1 = death, 0 = censored

# Fit Kaplan-Meier estimator
kmf = KaplanMeierFitter()
kmf.fit(durations=durations, event_observed=events, label="Kaplan-Meier")

# --- Greenwood standard errors -----
# event_table has index = times, columns: 'at_risk', 'observed', 'censored', ...
ev_tab = kmf.event_table

n = ev_tab["at_risk"].values            # n_i
```

```

d = ev_tab["observed"].values          #  $d_i$ 
S = kmf.survival_function_["Kaplan-Meier"].values #  $\hat{S}(t_j)$ 

# Greenwood sum: cumulative sum of  $d_i / [n_i (n_i - d_i)]$ 
term = np.zeros_like(d, dtype=float)
mask = (n > 0) & (n - d > 0)           # avoid division by zero at the very end
term[mask] = d[mask] / (n[mask] * (n[mask] - d[mask]))

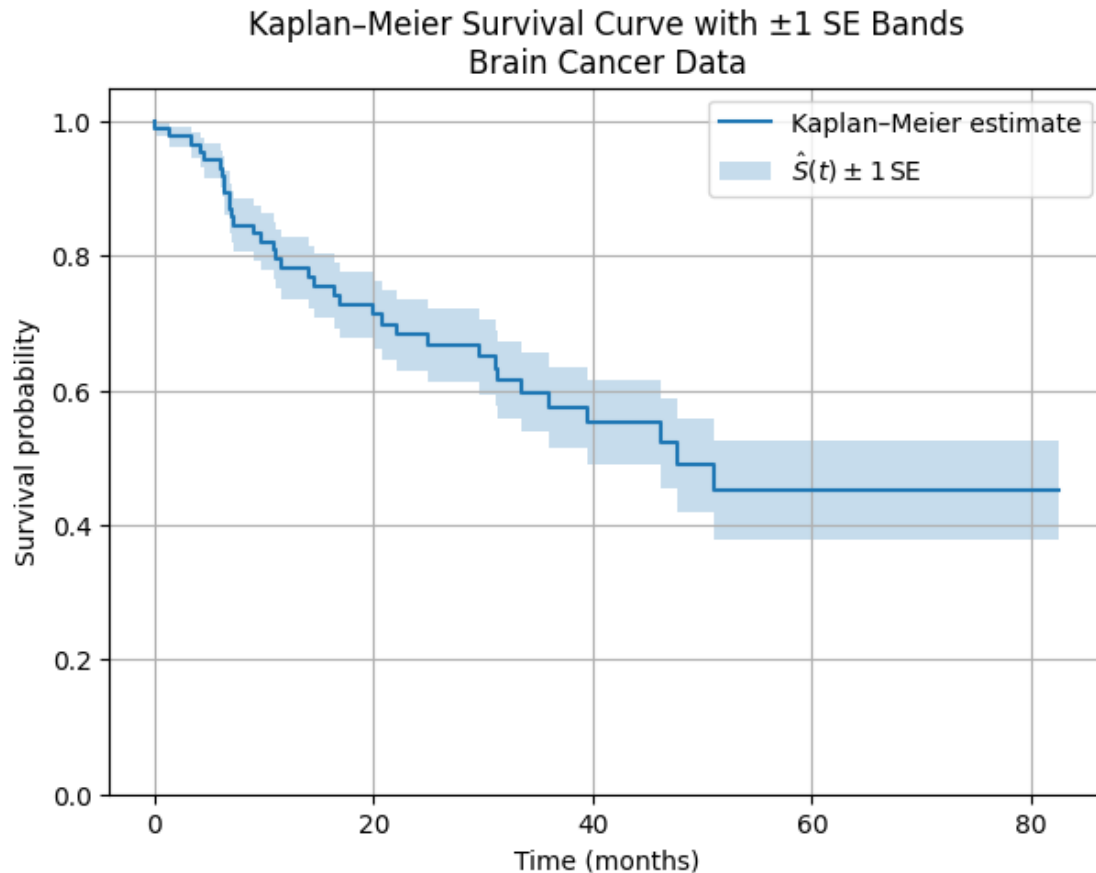
greenwood_sum = np.cumsum(term)
var_S = (S ** 2) * greenwood_sum
se_S = np.sqrt(var_S)

# Construct  $\pm 1$  SE bands, truncated to  $[0, 1]$ 
lower = np.clip(S - se_S, 0, 1)
upper = np.clip(S + se_S, 0, 1)
t = kmf.survival_function_.index

# --- Plot KM curve with  $\pm 1$  SE bands -----
plt.figure(figsize=(7, 5))
plt.step(t, S, where="post", label="Kaplan-Meier estimate")
plt.fill_between(t, lower, upper, step="post", alpha=0.25,
                 label=r" $\hat{S}(t) \pm 1 \cdot \mathrm{SE}$ ")

plt.xlabel("Time (months)")
plt.ylabel("Survival probability")
plt.ylim(0, 1.05)
plt.title("Kaplan-Meier Survival Curve with  $\pm 1$  SE Bands\nBrain Cancer Data")
plt.legend()
plt.grid(True)
plt.show()

```



- 1.2.3 (b) Draw a bootstrap sample of size $n = 88$ from the pairs (y_i, δ_i) , and compute the resulting Kaplan-Meier survival curve. Repeat this process $B = 200$ times. Use the results to obtain an estimate of the standard error of the Kaplan-Meier survival curve at each timepoint. Compare this to the standard errors obtained in (a).

```
[3]: # -----
# Original KM fit and Greenwood standard errors (from part a)
# -----
durations = BrainCancer["time"].values
events = BrainCancer["status"].values    # 1 = death, 0 = censored
n = len(BrainCancer)                    # should be 88

kmf = KaplanMeierFitter()
kmf.fit(durations, events)

# Time grid and KM estimate
t_grid = kmf.survival_function_.index.values
S_hat = kmf.survival_function_.iloc[:, 0].values
```

```

# Greenwood SE
ev_tab = kmf.event_table
n_at_risk = ev_tab["at_risk"].values
d_obs = ev_tab["observed"].values

term = np.zeros_like(d_obs, dtype=float)
mask = (n_at_risk > 0) & (n_at_risk - d_obs > 0)
term[mask] = d_obs[mask] / (n_at_risk[mask] * (n_at_risk[mask] - d_obs[mask]))

greenwood_sum = np.cumsum(term)
var_greenwood = (S_hat ** 2) * greenwood_sum
se_greenwood = np.sqrt(var_greenwood)

# -----
# Bootstrap estimate of the standard error of  $S(t)$ 
# -----
B = 200
boot_surv = np.empty((B, len(t_grid)))

rng = np.random.default_rng(seed=123) # fix seed for reproducibility

for b in range(B):
    # sample n pairs (time, status) with replacement
    idx = rng.integers(low=0, high=n, size=n)
    durations_b = durations[idx]
    events_b = events[idx]

    kmf_b = KaplanMeierFitter()
    kmf_b.fit(durations_b, events_b)

    # Evaluate the bootstrap KM curve on the original time grid
    boot_surv[b, :] = kmf_b.predict(t_grid)

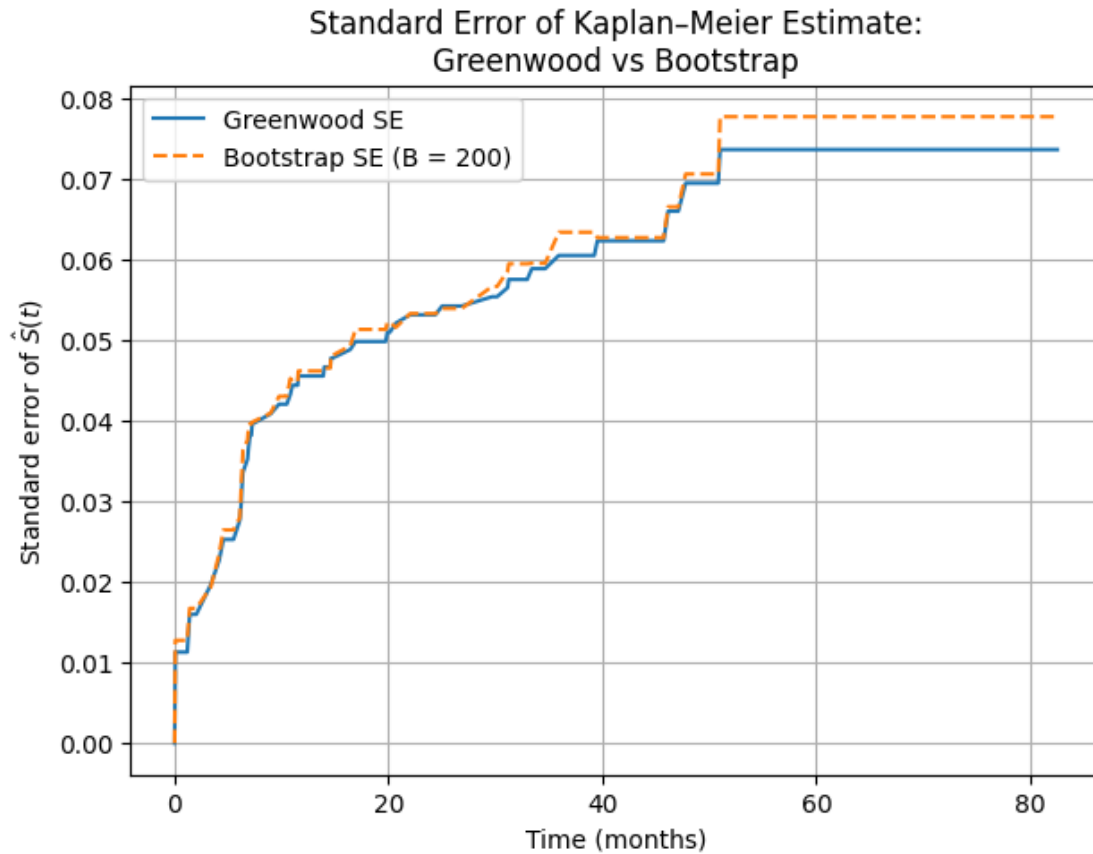
# Bootstrap SE at each time point:
#  $\widehat{\text{SE}}_{\text{boot}}(\hat{S}(t_j))$ 
se_boot = boot_surv.std(axis=0, ddof=1)

# -----
# Compare Greenwood vs bootstrap SEs
# -----
plt.figure(figsize=(7, 5))
plt.plot(t_grid, se_greenwood, label="Greenwood SE", linestyle="-")
plt.plot(t_grid, se_boot, label="Bootstrap SE (B = 200)", linestyle="--")

plt.xlabel("Time (months)")
plt.ylabel(r"Standard error of  $\hat{S}(t)$ ")

```

```
plt.title("Standard Error of Kaplan-Meier Estimate:\nGreenwood vs Bootstrap")
plt.legend()
plt.grid(True)
plt.show()
```



1.2.4 (c) Fit a Cox proportional hazards model that uses all of the predictors to predict survival. Summarize the main findings.

```
[10]: # Create design matrix: all predictors, with dummy variables for categoricals
X = BrainCancer.drop(columns=["time", "status"])
X = pd.get_dummies(X, drop_first=True)

# Combine with survival outcome
df_cox = pd.concat([BrainCancer[["time", "status"]], X], axis=1)

# Fit Cox proportional hazards model
cph = CoxPHFitter()
cph.fit(df_cox, duration_col="time", event_col="status")
```

```
# Show summary (coefficients, SE, z, p, etc.)
#cph.print_summary()
cph.summary
```

```
[10]:
```

	coef	exp(coef)	se(coef)	coef lower 95%	\
covariate					
ki	-0.061785	0.940085	0.066401	-0.191929	
gtv	0.036053	1.036711	0.022638	-0.008316	
ki_group	0.004556	1.004567	0.075340	-0.143107	
sex_Male	0.101372	1.106689	0.365738	-0.615461	
diagnosis_LG glioma	-1.088534	0.336710	0.575780	-2.217043	
diagnosis_Meningioma	-2.074319	0.125642	0.471239	-2.997930	
diagnosis_Other	-1.226068	0.293444	0.645976	-2.492157	
loc_Supratentorial	0.380002	1.462287	0.701360	-0.994639	
stereo_SRT	0.137370	1.147253	0.599667	-1.037956	

	coef upper 95%	exp(coef) lower 95%	\
covariate			
ki	0.068358	0.825366	
gtv	0.080422	0.991719	
ki_group	0.152220	0.866661	
sex_Male	0.818206	0.540392	
diagnosis_LG glioma	0.039974	0.108931	
diagnosis_Meningioma	-1.150708	0.049890	
diagnosis_Other	0.040022	0.082731	
loc_Supratentorial	1.754642	0.369857	
stereo_SRT	1.312696	0.354178	

	exp(coef) upper 95%	cmp to	z	p	\
covariate					
ki	1.070748	0.0	-0.930491	0.352117	
gtv	1.083744	0.0	1.592622	0.111245	
ki_group	1.164416	0.0	0.060477	0.951776	
sex_Male	2.266431	0.0	0.277172	0.781648	
diagnosis_LG glioma	1.040783	0.0	-1.890539	0.058686	
diagnosis_Meningioma	0.316413	0.0	-4.401842	0.000011	
diagnosis_Other	1.040834	0.0	-1.898008	0.057695	
loc_Supratentorial	5.781377	0.0	0.541807	0.587952	
stereo_SRT	3.716181	0.0	0.229078	0.818809	

	-log2(p)
covariate	
ki	1.505874
gtv	3.168189
ki_group	0.071306
sex_Male	0.355409
diagnosis_LG glioma	4.090841

diagnosis_Meningioma	16.507510
diagnosis_Other	4.115408
loc_Supratentorial	0.766231
stereo_SRT	0.288402

Solution : We fitted a Cox proportional hazards model with all predictors (`sex`, `diagnosis`, `loc`, `ki`, `gtv`, `stereo`) to the brain-cancer survival data (`time`, `status`).

- The global **likelihood ratio test** is highly significant ($\chi^2 = 39.5$ on 8 d.f., very small p -value), indicating that the covariates jointly have a strong association with survival. The model's **concordance index** is 0.79, showing good discriminative ability.
 - **Karnofsky index (ki)** has a negative coefficient ($\hat{\beta} = -0.06$, $p < 0.005$), with hazard ratio $\exp(\hat{\beta}) = 0.94$ (95% CI: 0.91–0.98). Each 1-point increase in `ki` multiplies the hazard by about 0.94, i.e., **better performance status is strongly associated with a lower instantaneous risk of death**.
 - **Diagnosis** is also strongly related to survival. The reference category is **high-grade glioma (HG glioma)**. Compared with HG glioma:
 - **Meningioma** has $\hat{\beta} = -2.08$, $\exp(\hat{\beta}) = 0.12$ (95% CI: 0.05–0.30, $p < 0.005$), meaning the hazard is about **88% lower** than for HG glioma.
 - **Other** diagnoses have $\exp(\hat{\beta}) = 0.29$ (95% CI: 0.09–0.99, $p \approx 0.05$), also indicating a substantially lower hazard.
 - **Low-grade glioma (LG glioma)** shows a reduced hazard ($\exp(\hat{\beta}) = 0.34$, 95% CI: 0.11–1.03, $p \approx 0.06$), with borderline statistical significance.
- Overall, patients with **HG glioma** have the worst prognosis, while those with **meningioma** (and to a lesser extent “other” and LG glioma) have much better survival.
- The remaining covariates show no clear evidence of association with survival after adjustment for the above:
 - **Tumor volume (gtv)** has a small positive effect ($\exp(\hat{\beta}) = 1.04$, 95% CI: 0.99–1.08, $p = 0.11$).
 - **Sex (Male vs Female)**, **tumor location** (supratentorial vs infratentorial), and **stereotactic method** (SRT vs SRS) have hazard ratios close to 1 with wide confidence intervals and large p -values ($p > 0.5$), suggesting **no strong independent effect** on survival in this dataset.

In summary, the Cox model indicates that survival is primarily driven by **diagnosis type** and **Karnofsky index**, with high-grade glioma and poor functional status associated with markedly worse outcomes, whereas sex, tumor location, stereotactic method, and tumor volume play at most a minor role once these variables are taken into account.

- 1.2.5 (d) Stratify the data by the value of k_i . (Since only one observation has $k_i == 40$, you can group that observation together with the observations that have $k_i == 60$.) Plot Kaplan-Meier survival curves for each of the five strata, adjusted for the other predictors.

```
[5]: # -----
# 1. Load data and create Karnofsky strata (5 levels)
#     Group ki == 40 together with ki == 60
# -----

BrainCancer["ki_group"] = BrainCancer["ki"].replace({40: 60})
ki_levels = np.sort(BrainCancer["ki_group"].unique()) # e.g. [60, 70, 80, 90, 100]

# -----
# 2. Fit a Cox model with all predictors, using numeric/dummy
#     covariates (this will be used to get adjusted survival curves)
# -----
# Keep original predictors, but drop the numeric ki (we now use ki_group)
predictors = ["sex", "diagnosis", "loc", "stereo", "gtv", "ki_group"]

df = BrainCancer[predictors + ["time", "status"]].copy()
df["ki_group"] = df["ki_group"].astype("category")

# Dummy-code categorical predictors (including ki_group)
X = pd.get_dummies(df.drop(columns=["time", "status"]), drop_first=True)
df_cox = pd.concat([df[["time", "status"]], X], axis=1)

cph = CoxPHFitter()
cph.fit(df_cox, duration_col="time", event_col="status")

# -----
# 3. Build representative covariate profiles:
#     - For each ki stratum, set ki_group to that value
#     - Other predictors set to mean (numeric) or mode (categorical)
# -----
rep = {
    "gtv": BrainCancer["gtv"].mean(),
    "sex": BrainCancer["sex"].mode()[0],
    "diagnosis": BrainCancer["diagnosis"].mode()[0],
    "loc": BrainCancer["loc"].mode()[0],
    "stereo": BrainCancer["stereo"].mode()[0],
}

profiles = []
index_names = []
```

```

for k in ki_levels:
    row = rep.copy()
    row["ki_group"] = k
    profiles.append(row)
    index_names.append(f"ki = {k}")

profiles = pd.DataFrame(profiles)
profiles["ki_group"] = profiles["ki_group"].astype("category")
profiles.index = index_names

# Dummy-code profiles using the SAME encoding as the Cox model
profiles_dummies = pd.get_dummies(profiles, drop_first=True)
profiles_dummies = profiles_dummies.reindex(
    columns=df_cox.drop(columns=["time", "status"]).columns,
    fill_value=0,
)

# -----
# 4. Predict survival curves for each ki stratum (adjusted for
#     the other predictors) and plot them
# -----
surv_curves = cph.predict_survival_function(profiles_dummies)

fig, ax = plt.subplots(figsize=(8, 6))
surv_curves.plot(ax=ax)

ax.set_xlabel("Time (months)")
ax.set_ylabel("Estimated survival probability")
ax.set_title("Adjusted survival curves by Karnofsky index strata")
ax.legend(title="Karnofsky strata")
ax.grid(True)

plt.show()

```

